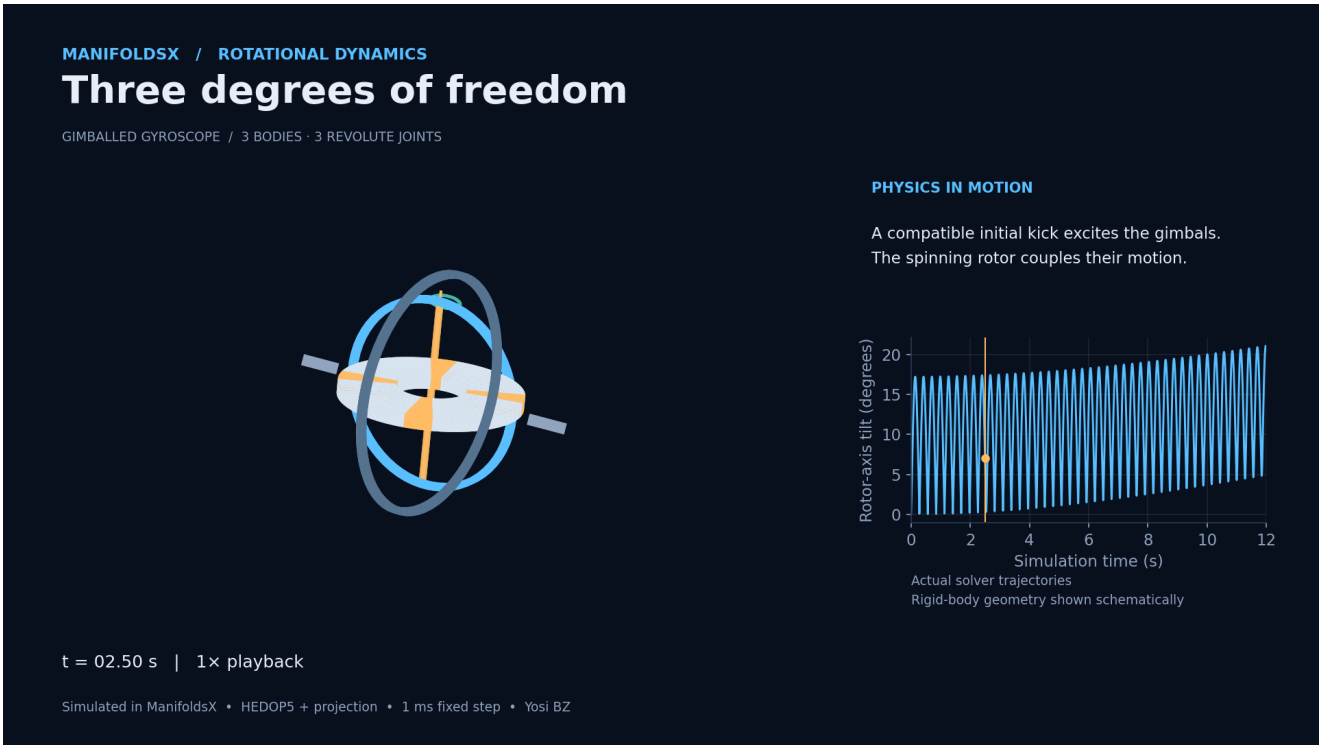
Three studies in rotational dynamics

Gimballed gyroscope, spinning top and torque-free tumbling

A spinning rotor couples the motion of two gimbals. A top precesses under gravity. A free body flips without an applied torque. We simulated each motion in ManifoldsX for 12 seconds and checked the results at three time steps.



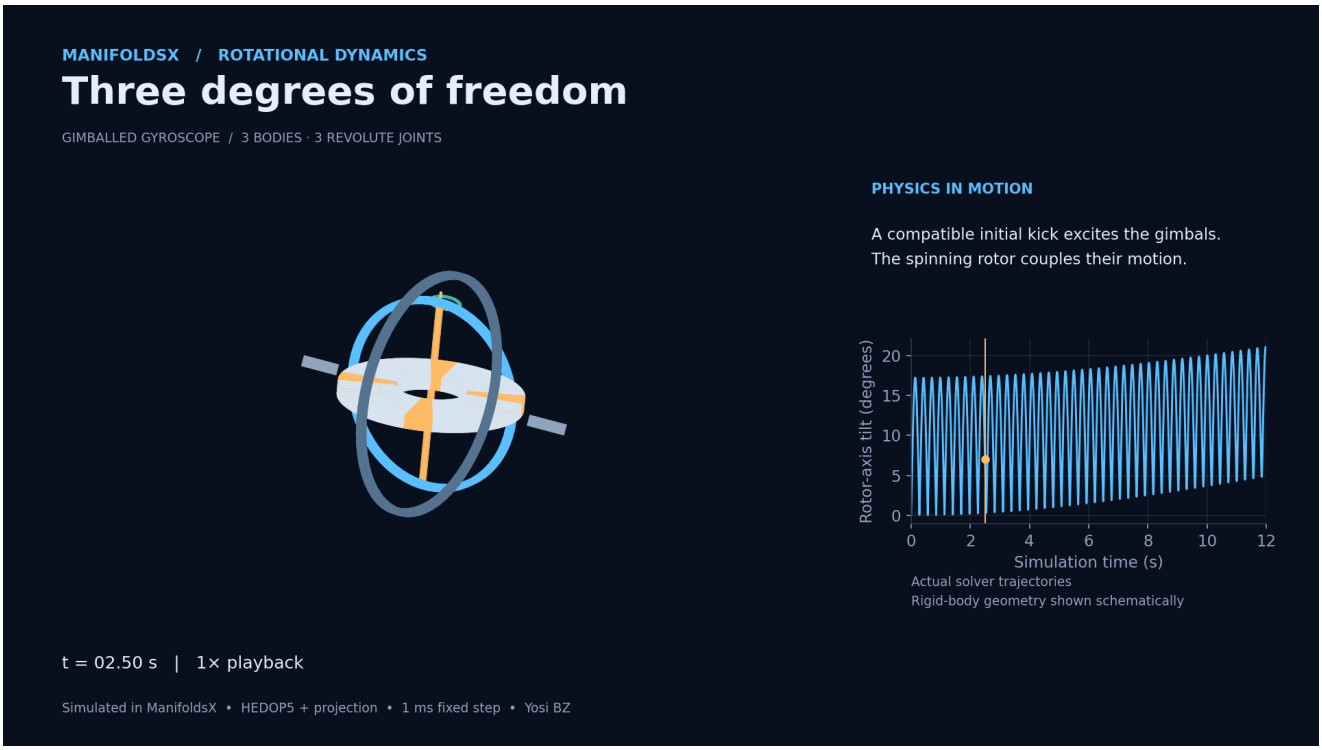
Study	Main result at a 1 ms step
3-DOF gimballed gyro	Halving the step changes the calculated orientation by at most 8.42e-11 rad. Energy and the two conserved momentum components remain nearly constant.
Spinning heavy top	The top follows the exact steady-precession solution, with a maximum orientation error of 3.68e-8 rad.
Asymmetric tumbling body	The body flips four times in 12 seconds. Its angular velocity agrees with the analytical solution to within 1.57e-9 rad/s.

The animations use the motion calculated by ManifoldsX. The following pages explain each model, its reference solution or conservation checks, and the measured errors.

01 / CONSTRAINED ROTATION

Three-DOF gimballed gyroscope

The rotor sits inside two freely moving gimbals. We start it spinning and give the whole assembly an initial rotation about the outer hinge. The rotor then couples the motion of the two gimbals, producing the oscillation shown below.



Model	Initial conditions
Each gimbal: mass 1 kg; I = diag(0.1, 0.1, 0.1) kg m^2. Rotor: mass 5 kg; I = diag(0.3, 0.3, 0.5) kg m^2.	Outer and inner angular velocity: (3, 0, 0) rad/s. Rotor angular velocity: (3, 0, 20) rad/s. Initial rotations: identity.

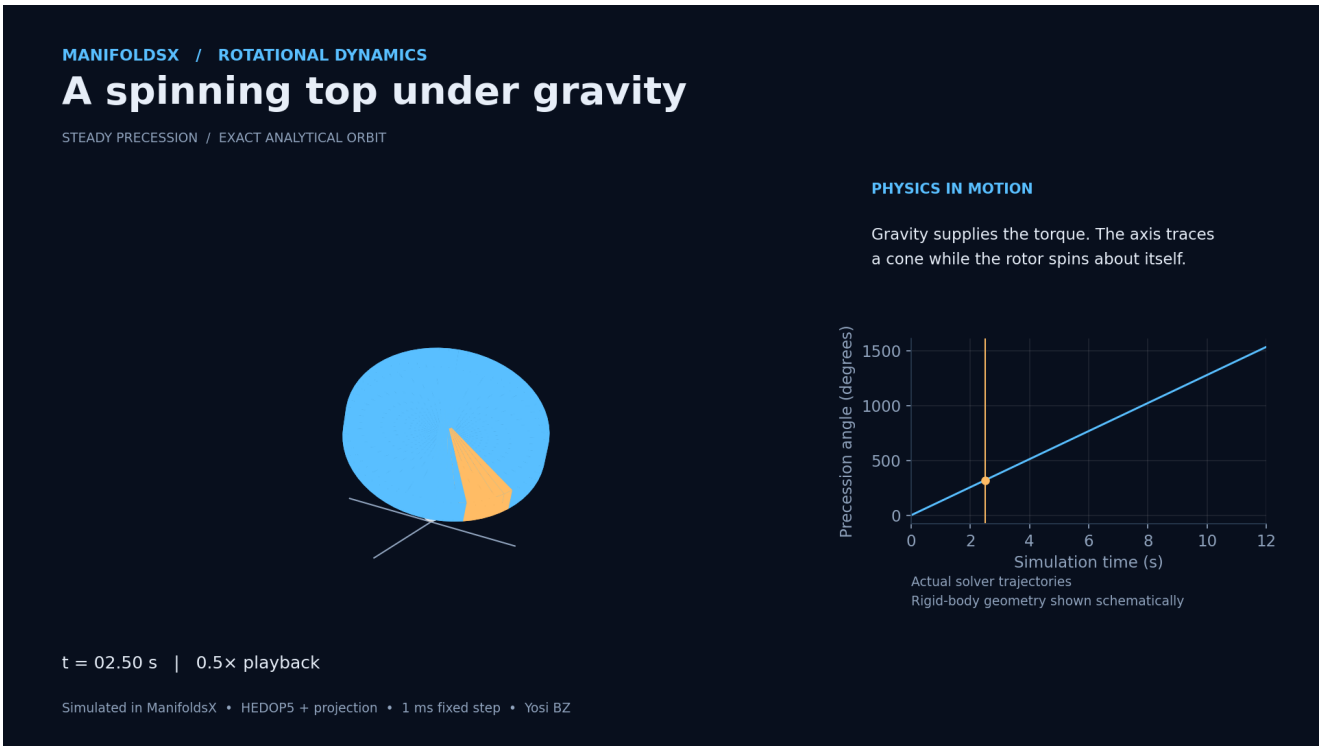
The three hinge axes initially point along X, Y and Z. All centres of mass lie at the pivot; gravity and bearing friction are disabled. The support can transmit torque, but angular momentum along its free X axis is conserved. Rotor axial spin and total energy should also stay constant.

Measured over 12 s at 1 ms	Peak change
Relative mechanical energy	1.23e-12
Total X angular momentum (kg m^2/s)	2.44e-11
Rotor axial angular speed (rad/s)	2.26e-11

Halving the time step changes the orientation of any body by at most 8.42e-11 rad. Together with the conservation checks, this shows that the calculated motion is insensitive to further step refinement at this scale.

Spinning heavy top

This top spins about its own axis while that axis travels around the vertical. We choose the initial motion for steady precession, so the tilt remains constant. That gives us an exact solution against which to check both the direction of the axis and the full body orientation.



Model	Exact motion
Mass: 1 kg; pivot-to-COM distance: 0.2 m. COM principal inertias: (0.02, 0.02, 0.01) kg m^2. Initial tilt: 0.45 rad; gravity: 9.81 m/s^2.	Axial spin: 100 rad/s. Precession rate: 2.230883 rad/s. Tilt remains 0.45 rad. Video: first 6 simulated seconds at half speed.

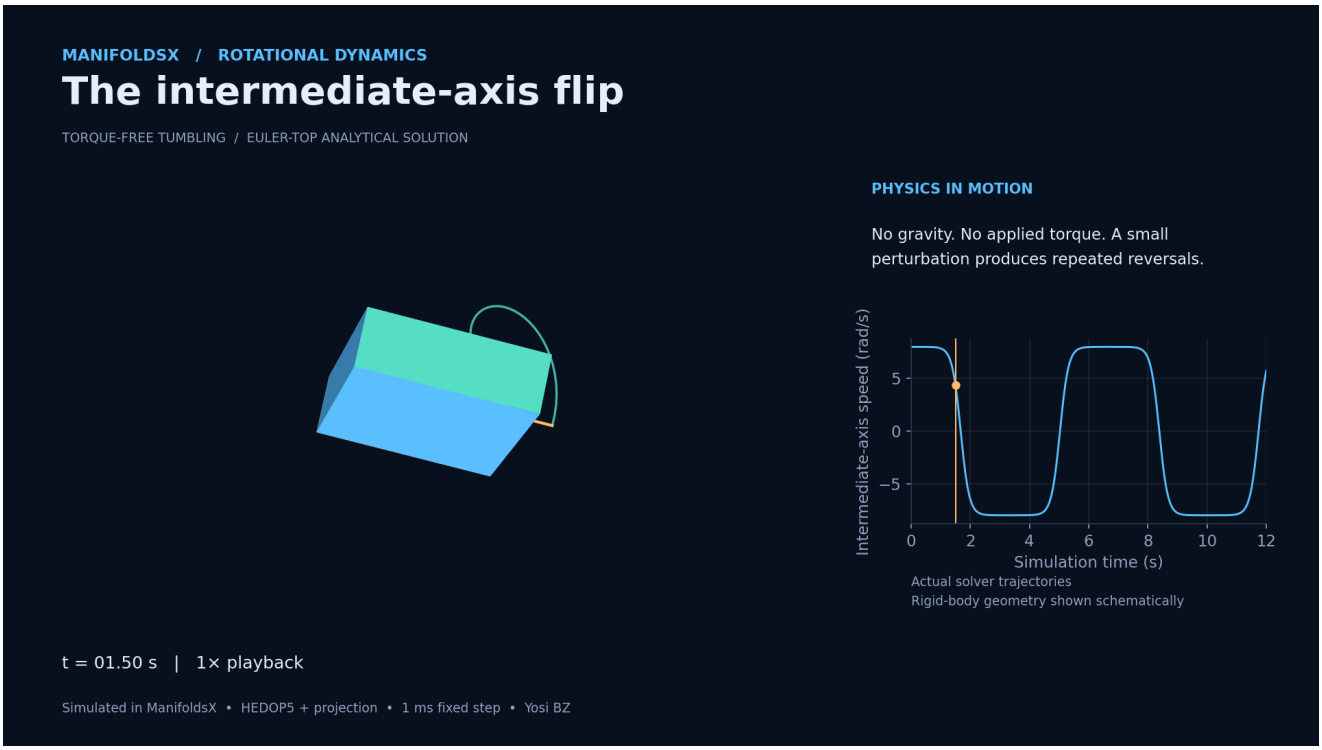
Gravity acts at the centre of mass, 0.2 m from the pivot, and supplies the torque that produces precession. The reference solution combines rotation about the vertical with spin about the inclined axis. Its equations are included in the accompanying source notes.

Analytical comparison at 1 ms	Peak error
Complete orientation (rad)	3.68e-08
Symmetry-axis direction (rad)	4.28e-09
Relative mechanical energy change	1.39e-11

The maximum orientation error falls from 3.68e-8 rad at 1 ms to 3.80e-11 rad at 0.25 ms. Each halving of the step reduces the error by about 31 times, consistent with fifth-order convergence.

Asymmetric tumbling body

The body starts spinning almost entirely about the axis with the middle principal moment of inertia. This rotation is unstable: a small initial disturbance grows until the body flips. The pattern repeats even though no external torque acts.



Model	Analytical reference
Mass: 1 kg. Principal inertias: (0.2, 0.31, 0.47) kg m^2. Initial body angular velocity: (0, 8, 0.05) rad/s. Zero gravity; no joints or applied forces.	The exact Euler-top solution gives all three angular-velocity components using Jacobi elliptic functions. We evaluate it at 45-digit precision and compare it with the simulation every 5 ms.

Measured over 12 s at 1 ms	Result
Intermediate-axis speed reversals	4
Angular-velocity vector error (rad/s)	1.57e-09
Relative angular-momentum vector error	2.67e-14
Relative rotational-energy change	4.97e-14

The simulation captures four reversals of the intermediate-axis angular velocity. Energy and world angular momentum remain nearly constant throughout. The animated cuboid has the same principal inertias as the simulated body.

How the results were checked

Each case was run at fixed steps of 1, 0.5 and 0.25 ms. All nine runs reached 12 seconds without retries or cutbacks. We checked energy and momentum at every step, the top orientation at every step, and the tumbling angular velocity every 5 ms.

Step (ms)	Top attitude error (rad)	Tumble speed error (rad/s)
1.0	3.68e-08	1.57e-09
0.5	1.17e-09	1.31e-09
0.25	3.8e-11	4.9e-09

The top shows a clear reduction in error as the step is refined. The tumbling case is already within a few billionths of a radian per second of the analytical solution; smaller steps do not consistently improve that result.

Gimbal refinement	Maximum orientation difference (rad)
1 vs 0.5 ms	8.42e-11
0.5 vs 0.25 ms	5.15e-12

Solver settings. All runs use HEDOP5 with position and velocity projection. The projection triggers and tolerances are 1e-13 for position and 1e-12 for velocity, with up to 10 position-projection iterations. Steps remain fixed throughout. The solver uses double precision; the analytical tumbling reference uses 45-digit arithmetic.

Source. ManifoldsX/MBD commit 70ef1ffa42bd8a93a716845534b4f8d77ec7a0eb. The accompanying archive contains the model driver, engine snapshot, analytical calculations, raw results and scripts used to produce the figures and report.

Reproducing the study. The archive README gives the build commands, model origins and reference equations. Gimbal comparisons use the maximum orientation difference across all three bodies. Top errors measure the relative rotation from the analytical solution; tumbling errors measure the three-component angular-velocity difference.

References

1. J. Peraire and S. Widnall, MIT 16.07, Lecture L30: 3D Rigid Body Dynamics: Tops and Gyroscopes (2009). MIT OpenCourseWare lecture and derivation

2. NIST Digital Library of Mathematical Functions, Section 22.13, derivatives of Jacobi elliptic functions. dlmf.nist.gov/22.13

The gimbal and top shapes in the animations are schematic. Their simulated masses and inertias are listed with each case.